Daniel Peelen

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Education

University of Pittsburgh September 2023 - May 2027

Bachelor of Science in Computer Science, Minor in Hungarian

- **GPA:** 3.9/4.0 (Dean's List)
- **Coursework:** Intro to Deep Learning, Intro to ML, Data Structures & Algorithms, Probability Theory

Work Experience

Software Engineering Intern May 2026 – August 2026

Microsoft *Redmond, WA*

- Analyzed 1,000,000+ Word Copilot crashes with Kusto and OpenTelemetry by building an LLM reliability agent.
- Shipped an **org-wide plug-in for Microsoft's internal Copilot**, giving engineers on-demand crash analyses.
- Fixed Word bugs surfaced by agent's output, **eliminated 50,000+ enterprise-facing crashes** over 2 releases.

Machine Learning Engineer Intern January 2026 – April 2026

Synopsys *Pittsburgh, PA*

- Designed and implemented Ansys Mechanical's RAG coding agent memory solution with semantic clustering.
- Built chatbot's full test suite and a multi-turn eval harness with bucket visualization on Ansys test infrastructure.
- **Raised multi-turn accuracy from 40% to 82%** by shipping Go code to production behind feature flags.

Machine Learning Engineer Intern June 2025 – August 2025

Thursday *Pittsburgh, PA*

- **Carnegie Mellon startup;** Shipped batch fashion recommendations in under 200ms to **10K monthly users**.
- Served real-time recommendations with a custom DeepFM model using Kafka event streaming and Redis caching.

Teaching Assistant August 2024 – April 2025

University of Pittsburgh *Pittsburgh, PA*

- Led recitations and practices for 100+ students in Discrete Math, **raising average final scores from 78 to 84**
- Held 5+ hours of weekly office hours, coaching students on proof techniques and algorithmic thinking.

Projects

DannyTorch | *Python, NumPy*

- Built an **ML framework from scratch with NumPy with no AI**; built full autograd over compute graphs.
- **Coded 10+ transformer parts** including attention, multi-layered perceptron, positional encodings, and more.
- Validated gradient correctness by training LM, reducing **loss from 4.9 to 1.2** over 500 epochs on a Colab A100.

AnkiMate | *Python, Flan-T5, PyTorch, Flask, React*

- Fine-tuned Flan-T5 on self-curated 10K-row dataset to make cards from PDF/PPT, reduced validation loss 58%.
- Designed and deployed a **Flask backend for flashcard generation**; used React and Tailwind for a responsive UI.

Mantis | *Python, YOLOv8, OpenCV, GCP, Docker* **2nd place, CalHacks 2025**

- Built a full-stack surveillance platform detecting theft and weapons in live feeds with real-time owner alerts
- Architected a Python detection core streaming over WebSockets into a reactive Next.js/Convex dashboard on GCP
- Processed 4 concurrent feeds at **30 FPS under 350ms** and clipped 10+ hr footage, **placing 2nd of 400+ teams**

Skills

Languages: Python, C/C++, Go, Java, JavaScript, TypeScript, HTML/CSS, OCaml, SQL
Frameworks: PyTorch, HuggingFace Transformers, Scikit-learn, NumPy, Pandas, OpenCV, FastAPI, Flask
Infra & Tools: LLMs, Git, CLI, Docker, AWS, Kafka, Redis, GCP, Hadoop, Linux
Concepts: Machine Learning, AI, Model Evaluation, RAG, Distributed Training, Fine-tuning, RLHF

Extracurriculars

Director, SteelHacks January 2025 – September 2025

- Led team to grow event by **21%** from year prior to 600+ attendees, raised $50K+ from Google, Amazon, and PNC.

President, Quiz Bowl September 2024 – Present

- Placed 4th individually at nationals; led team to 3rd place finish at Collegiate Nationals for first time in 20 years.